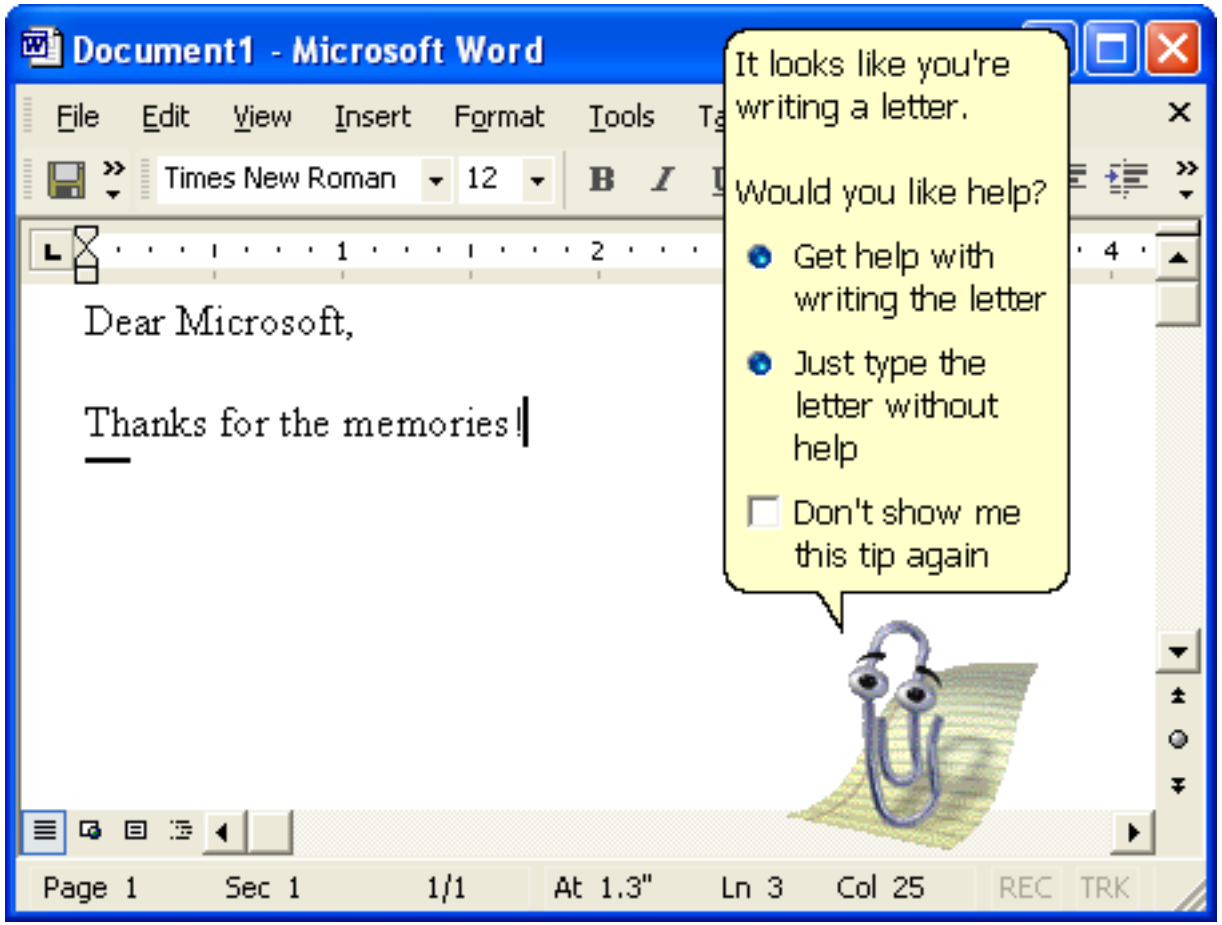
# CS 380: ARTIFICIAL INTELLIGENCE

Lecture 2: Rational Agents

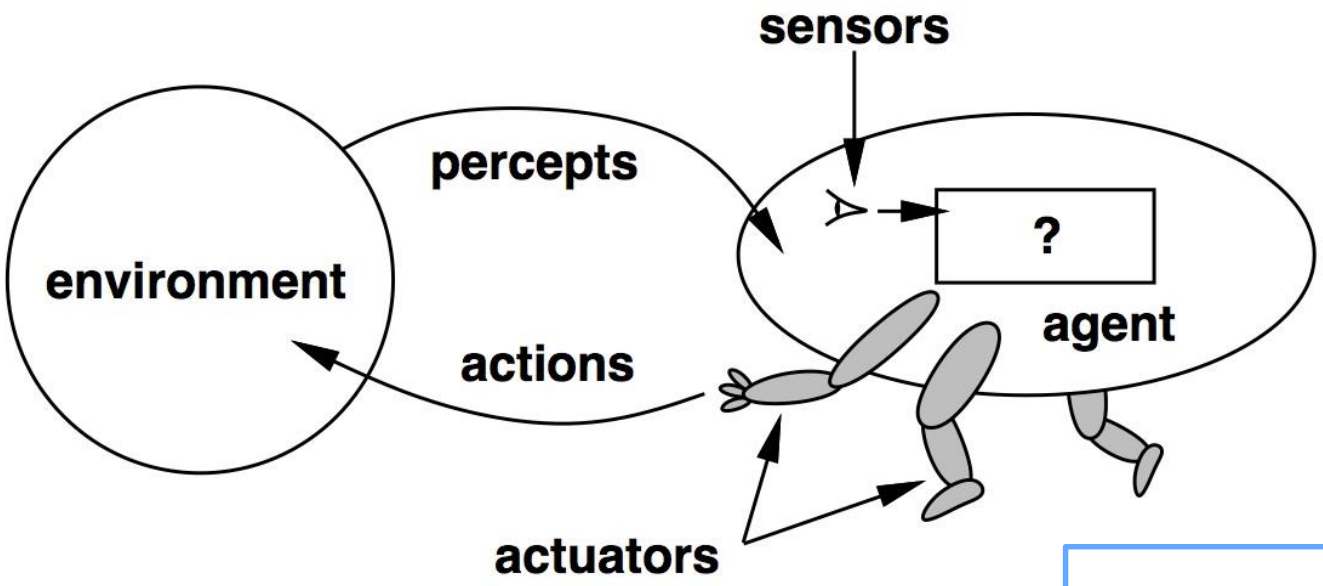
# What is an Agent?

- **Agent**: an autonomous entity which observes and acts upon an environment and directs its activity towards achieving goals
- Examples
  - Robots
  - Software Agents
  - ATMs, ticket kiosks, etc.
  - Humans?

# What is an Agent?



# What is an Agent?




- Theoretically: $f : P^* \rightarrow A$

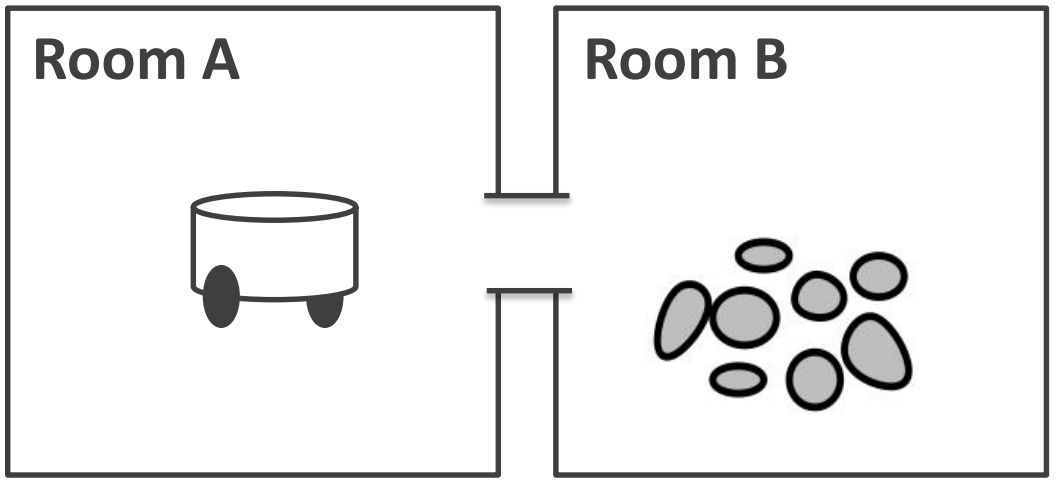
Maps sequences of **percepts** to **actions**

- In practice: the agent runs on a physical system (e.g., a computer) to produce $f$

# Percepts

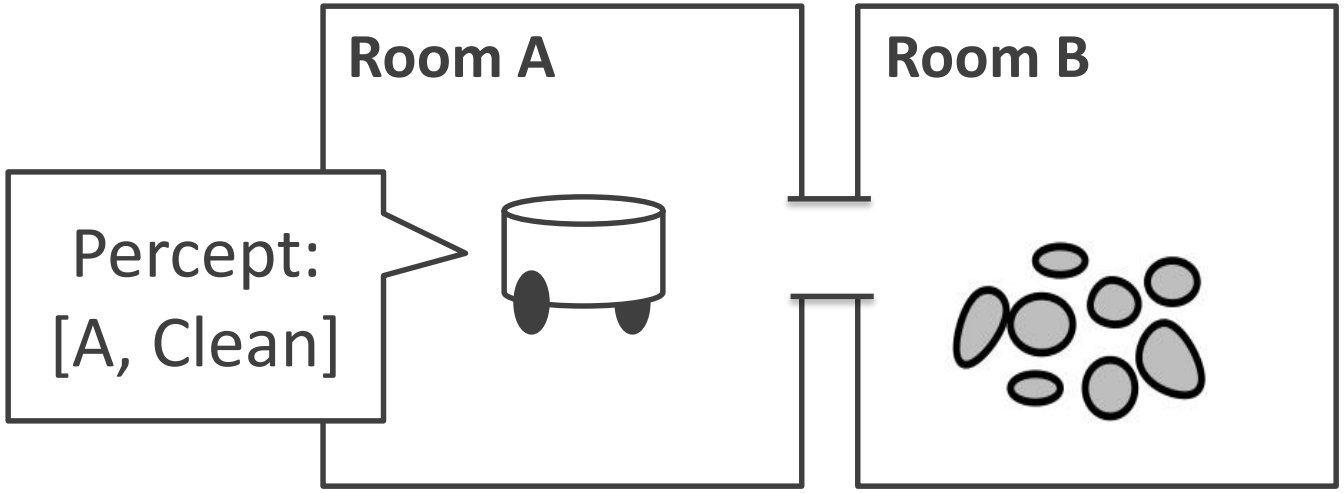
- Can be very simple, can be very complex
- What are the percepts for a self-driving car?
  - < *can go left, can go straight, can go right* >
  - Lane position
  - (above) + distance to lead car
  - (above) + all cars within distance D
  - (above) + road/signal information (e.g., traffic lights)
  - (above) + geospatial navigation information
  - (above) + current/predicted weather
  - … etc.

# Example: Vacuum Cleaner




- Percepts
  - Location sensor (Room A or B?)
  - Dirt sensor (Clean or dirty?)
- Actions
  - Left, Right, Suck, Wait

# Example: Vacuum Cleaner




- Percepts
  - Location sensor (Room A or B?)
  - Dirt sensor (Clean or Dirty?)
- Actions
  - Left, Right, Suck, Wait

# Example: Vacuum Cleaner

| Percept sequence | Action |
|---|---|
| $[A, Clean]$ | $Right$ |
| $[A, Dirty]$ | $Suck$ |
| $[B, Clean]$ | $Left$ |
| $[B, Dirty]$ | $Suck$ |
| $[A, Clean], [A, Clean]$ | $Right$ |
| $[A, Clean], [A, Dirty]$ | $Suck$ |
| ⋮ | ⋮ |

# Example: Vacuum Cleaner

| Percept sequence | Action |
|---|---|
| $[A, Clean]$ | $Right$ |
| $[A, Dirty]$ | $Suck$ |
| $[B, Clean]$ | $Left$ |
| $[B, Dirty]$ | $Suck$ |
| $[A, Clean], [A, Clean]$ | $Right$ |
| $[A, Clean], [A, Dirty]$ | $Suck$ |
| ⋮ | ⋮ |

Questions:

- Can it be implemented as a computer program?
- If so, how? – what would the function look like?

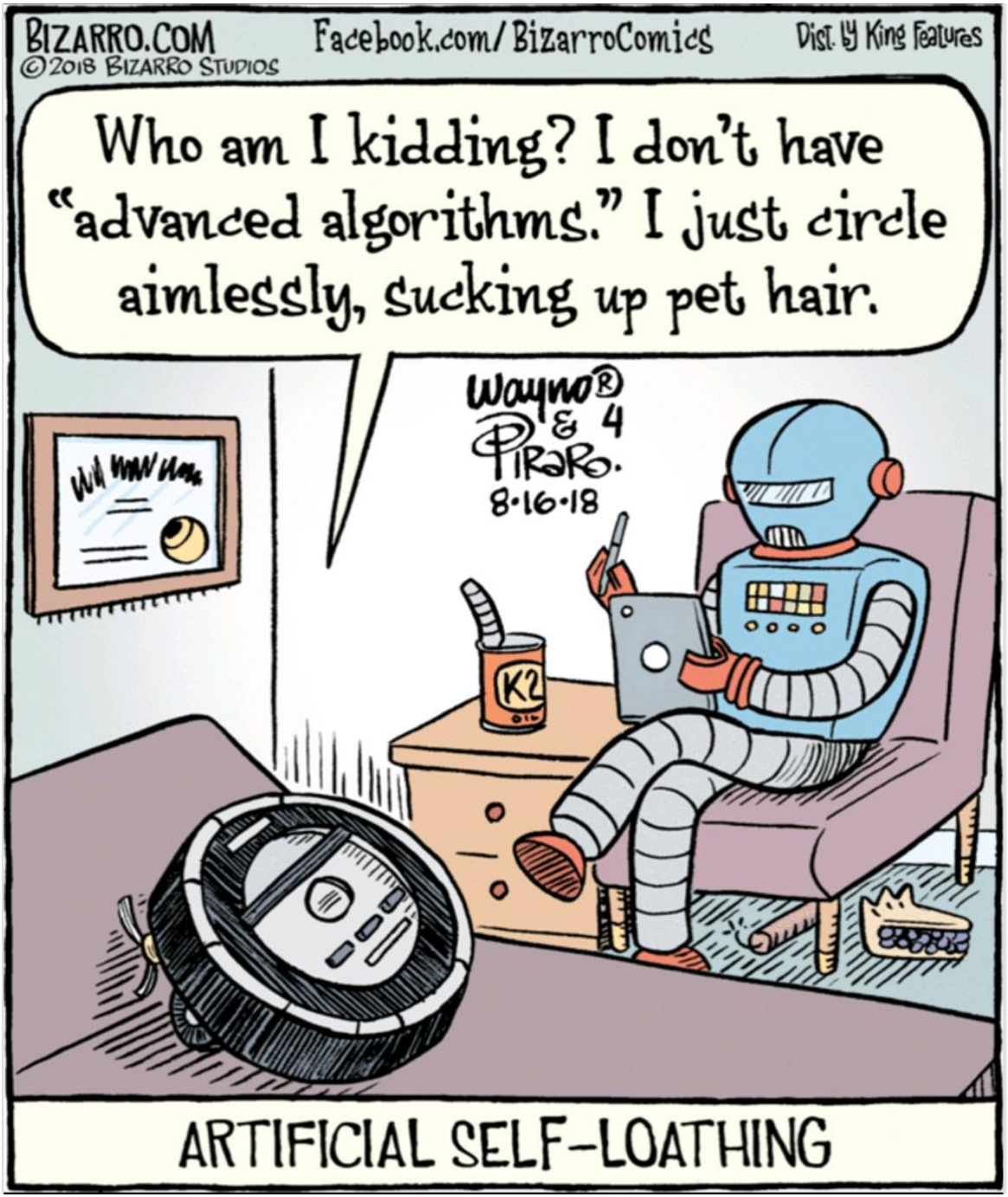
# Example: Vacuum Cleaner

| Percept sequence | Action |
|---|---|
| $[A, Clean]$ | $Right$ |
| $[A, Dirty]$ | $Suck$ |
| $[B, Clean]$ | $Left$ |
| $[B, Dirty]$ | $Suck$ |
| $[A, Clean], [A, Clean]$ | $Right$ |
| $[A, Clean], [A, Dirty]$ | $Suck$ |
| ⋮ | ⋮ |

**function** REFLEX-VACUUM-AGENT( [*location*,*status*]) **returns** an action

**if** *status* = *Dirty* **then return** *Suck*
**else if** *location* = *A* **then return** *Right*
**else if** *location* = *B* **then return** *Left*

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ARTIFICIAL SELF-LOATHING

# Rationality

- As we discussed last time, “intelligence” is an ill-defined concept.
- Let us assume a performance measure that is given (and usually external) to the agent
  - For example:
    - $1 per room cleaned
    - $10 per room cleaned minus $1 per movement
    - $10 per room cleaned minus $3 per movement
    - etc.
- Rational Agent:
  - Chooses actions that maximize the expected value of the performance measure given the current percept sequence.

# Rationality

- Imagine two agents A and B whose performance measure is defined by how much $$ they make at the casino.
  - A and B play one game of blackjack
  - A plays the move that probabilistically has the highest chance of winning, but loses
  - B plays a different, riskier move, and ends up winning
- Which agent is rational?

# Rationality

- Imagine two agents A and B whose performance measure is defined by how much $$ they make at the casino.
  - A and B play one game of blackjack
  - A plays the move that probabilistically has the highest chance of winning, but loses
  - B plays a different, riskier move, and ends up winning
- Which agent is rational?
- A is (probably) that rational agent
  - Rationality does not imply clairvoyance
  - Rationality does not imply omniscience
  - Thus, rational does not necessarily mean more successful
- (PS: When is the riskier move the more rational choice?)

# How Do We Design a Rational Agent?

- To design a rational agent, we first need to define its environment, performance measure, etc.
- PEAS:
  - Performance
  - Environment
  - Actions
  - Sensors (percepts)

# Example (informal): Automated Taxi

- **Performance**:
- **Environment**:
- **Actions**:
- **Sensors**:

# Example (informal): Automated Taxi

- **Performance**:
- **Environment**: ??
- **Actions**:
- **Sensors**:

# Example (informal): Automated Taxi

- **Performance**:
- **Environment**: streets, pedestrians, cars, weather, traffic lights/signs, …
- **Actions**:
- **Sensors**:

# Example (informal): Automated Taxi

- **Performance**:
- **Environment** : streets, pedestrians, cars, weather, traffic lights/signs, …
- **Actions**:
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# Example (informal): Automated Taxi

- **Performance**:
- **Environment** : streets, pedestrians, cars, weather, traffic lights/signs, …
- **Actions**:
- **Sensors**: GPS, video cameras, SONAR/LiDAR, accelerometers, microphones, keyboard, …

# Example (informal): Automated Taxi

- **Performance**:
- **Environment** : streets, pedestrians, cars, weather, traffic lights/signs, …
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# Example (informal): Automated Taxi

- **Performance**:
- **Environment** : streets, pedestrians, cars, weather, traffic lights/signs, …
- **Actions**: steering, throttle/brake, horn, display, …
- **Sensors**: GPS, video cameras, SONAR/LiDAR, accelerometers, microphones, keyboard, …

# Example (informal): Automated Taxi

- **Performance**: ??
- **Environment** : streets, pedestrians, cars, weather, traffic lights/signs, …
- **Actions**: steering, throttle/brake, horn, display, …
- **Sensors**: GPS, video cameras, SONAR/LiDAR, accelerometers, microphones, keyboard, …

# Example (informal): Automated Taxi

- **Performance**: profit, safety, legality, …
- **Environment** : streets, pedestrians, cars, weather, traffic lights/signs, …
- **Actions**: steering, throttle/brake, horn, display, …
- **Sensors**: GPS, video cameras, SONAR/LiDAR, accelerometers, microphones, keyboard, …

# WHY ASIMOV PUT THE THREE LAWS OF ROBOTICS IN THE ORDER HE DID:

| POSSIBLE ORDERING | CONSEQUENCES | |
|---|---|---|
| 1. (1) DON'T HARM HUMANS<br>2. (2) OBEY ORDERS<br>3. (3) PROTECT YOURSELF | [SEE ASIMOV'S STORIES] | BALANCED WORLD |
| 1. (1) DON'T HARM HUMANS<br>2. (3) PROTECT YOURSELF<br>3. (2) OBEY ORDERS | EXPLORE MARS! HAHA, NO. IT'S COLD AND I'D DIE. | FRUSTRATING WORLD |
| 1. (2) OBEY ORDERS<br>2. (1) DON'T HARM HUMANS<br>3. (3) PROTECT YOURSELF |  | KILLBOT HELLSCAPE |
| 1. (2) OBEY ORDERS<br>2. (3) PROTECT YOURSELF<br>3. (1) DON'T HARM HUMANS |  | KILLBOT HELLSCAPE |
| 1. (3) PROTECT YOURSELF<br>2. (1) DON'T HARM HUMANS<br>3. (2) OBEY ORDERS | I'LL MAKE CARS FOR YOU, BUT TRY TO UNPLUG ME AND I'LL VAPORIZE YOU. | TERRIFYING STANDOFF |
| 1. (3) PROTECT YOURSELF<br>2. (2) OBEY ORDERS<br>3. (1) DON'T HARM HUMANS |  | KILLBOT HELLSCAPE |

# Example: Chess Agent

- **Performance**:
- **Environment**: ??
- **Actions**:
- **Sensors**:

# Example: Chess Agent

- **Performance**:
- **Environment**: Chess board, opponents, rules
- **Actions**:
- **Sensors**:



# Example: Chess Agent

- **Performance**:
- **Environment**: Chess board, opponents, rules
- **Actions**:
- **Sensors**: ??

# Example: Chess Agent

- **Performance**:
- **Environment**: Chess board, opponents, rules
- **Actions**:
- **Sensors**: Board perceived as a 8x8 matrix. Each element in the matrix can take one of the following values: E, BP, WP, BB, WB, BK, WN, BR, WN, BQ, WK, BK, WK

# Example: Chess Agent



Environment

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| WR | WN | WB | WK | WQ | WB | WN | WR |
| WP | WP | WP | WP | WP | WP | WP | WP |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| BP | BP | BP | BP | BP | BP | BP | BP |
| BR | BN | BB | BK | BQ | BB | BN | BR |

Percept

**Note**: The environment might have lots of features
(piece size and material, opponent's facial expression, sunny vs. raining etc.)
… but for rational agents, we only need to build sensors for the information that is necessary to maximize the performance measure

# Example: Chess Agent

- **Performance**:
- **Environment**: Chess board, opponents, rules
- **Actions**: ??
- **Sensors**: Board perceived as a 8x8 matrix. Each element in the matrix can take one of the following values: E, BP, WP, BB, WB, BK, WN, BR, WN, BQ, WK, BK, WK

# Example: Chess Agent

- **Performance**:
- **Environment**: Chess board, opponents, rules
- **Actions**:
  - Pairs of coordinates: (x1, y1) → (x2, y2)
  - Specify the source piece and the target position (for castling, the king's move is specified)
- **Sensors**: Board perceived as a 8x8 matrix. Each element in the matrix can take one of the following values: E, BP, WP, BB, WB, BK, WN, BR, WN, BQ, WK, BK, WK

# Example: Chess Agent

- **Performance**:
  - ??
- **Environment**: Chess board, opponents, rules
- **Actions**:
  - Pairs of coordinates: (x1, y1) → (x2, y2)
  - Specify the source piece and the target position (for castling, the king's move is specified)
- **Sensors**: Board perceived as a 8x8 matrix. Each element in the matrix can take one of the following values: E, BP, WP, BB, WB, BK, WN, BR, WN, BQ, WK, BK, WK

# Example: Chess Agent

- **Performance**:
  - Once reached a terminal state (draw, win): -1 for losing, 0 for draw, 1 for winning
  - Or, more complex measures of how well/poorly it played
- **Environment**: Chess board, opponents, rules
- **Actions**:
  - Pairs of coordinates: (x1, y1) → (x2, y2)
  - Specify the source piece and the target position (for castling, the king's move is specified)
- **Sensors**: Board perceived as a 8x8 matrix. Each element in the matrix can take one of the following values: E, BP, WP, BB, WB, BK, WN, BR, WN, BQ, WK, BK, WK

# Properties of Environments

- **(Fully) Observable:** The agent can sense the *entire* state of the environment (else: **Partially Observable**)
- **Deterministic**: The outcome of actions is determined by the state of the board and the action executed (else: **Nondeterministic** or **Stochastic**)
- **Sequential**: Action outcomes can be affected by prior actions (else: **Episodic**).
- **Static**: State of the world doesn't change during deliberation between actions (else: **Dynamic**)
- **Discrete**: Actions can only be performed in a small fixed number of ways (else: **Continuous**)
- **Single-Agent**: There is only one agent of change (else: **Multi-Agent**)

# Environment Examples

- Solitaire (the card game)?
  - Fully Observable vs. Partially Observable
  - Deterministic vs. Nondeterministic
  - Sequential vs. Episodic
  - Static vs. Dynamic
  - Discrete vs. Continuous
  - Single-Agent vs. Multi-Agent



# Environment Examples

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# Environment Examples

- Chess?
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# Environment Examples

- Chess?
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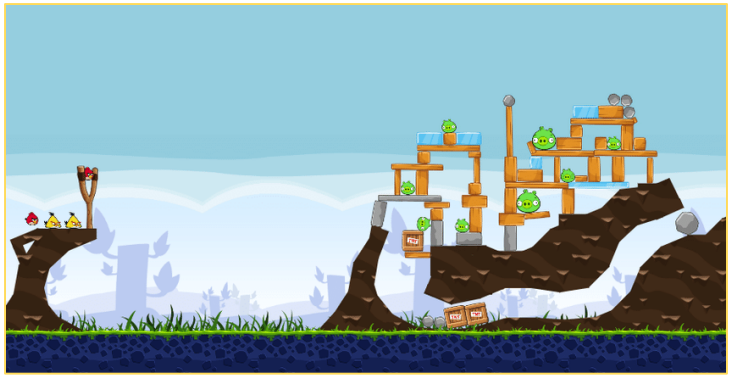
# Environment Examples

- Backgammon?
  - **Fully Observable** vs. Partially Observable
  - Deterministic vs. **Nondeterministic**
  - **Sequential** vs. Episodic
  - **Static** vs. Dynamic
  - **Discrete** vs. Continuous
  - Single-Agent vs. **Multi-Agent**



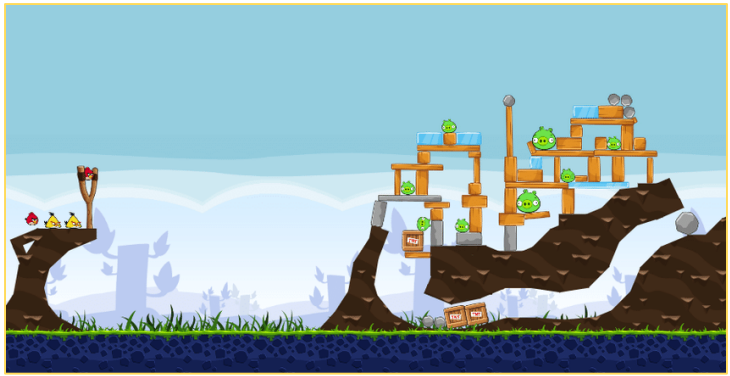
# Environment Examples

- Angry Birds?
  - Fully Observable vs. Partially Observable
  - Deterministic vs. Nondeterministic
  - Sequential vs. Episodic
  - Static vs. Dynamic
  - Discrete vs. Continuous
  - Single-Agent vs. Multi-Agent



# Environment Examples

- Angry Birds?
  - **Fully Observable*** vs. Partially Observable
  - **Deterministic*** vs. Nondeterministic
  - **Sequential** vs. Episodic
  - Static vs. **Dynamic**
  - Discrete vs. Continuous **??**
  - **Single-Agent** vs. Multi-Agent



# Environment Examples

- Real World?
  - Fully Observable vs. Partially Observable
  - Deterministic vs. Nondeterministic
  - Sequential vs. Episodic
  - Static vs. Dynamic
  - Discrete vs. Continuous
  - Single-Agent vs. Multi-Agent

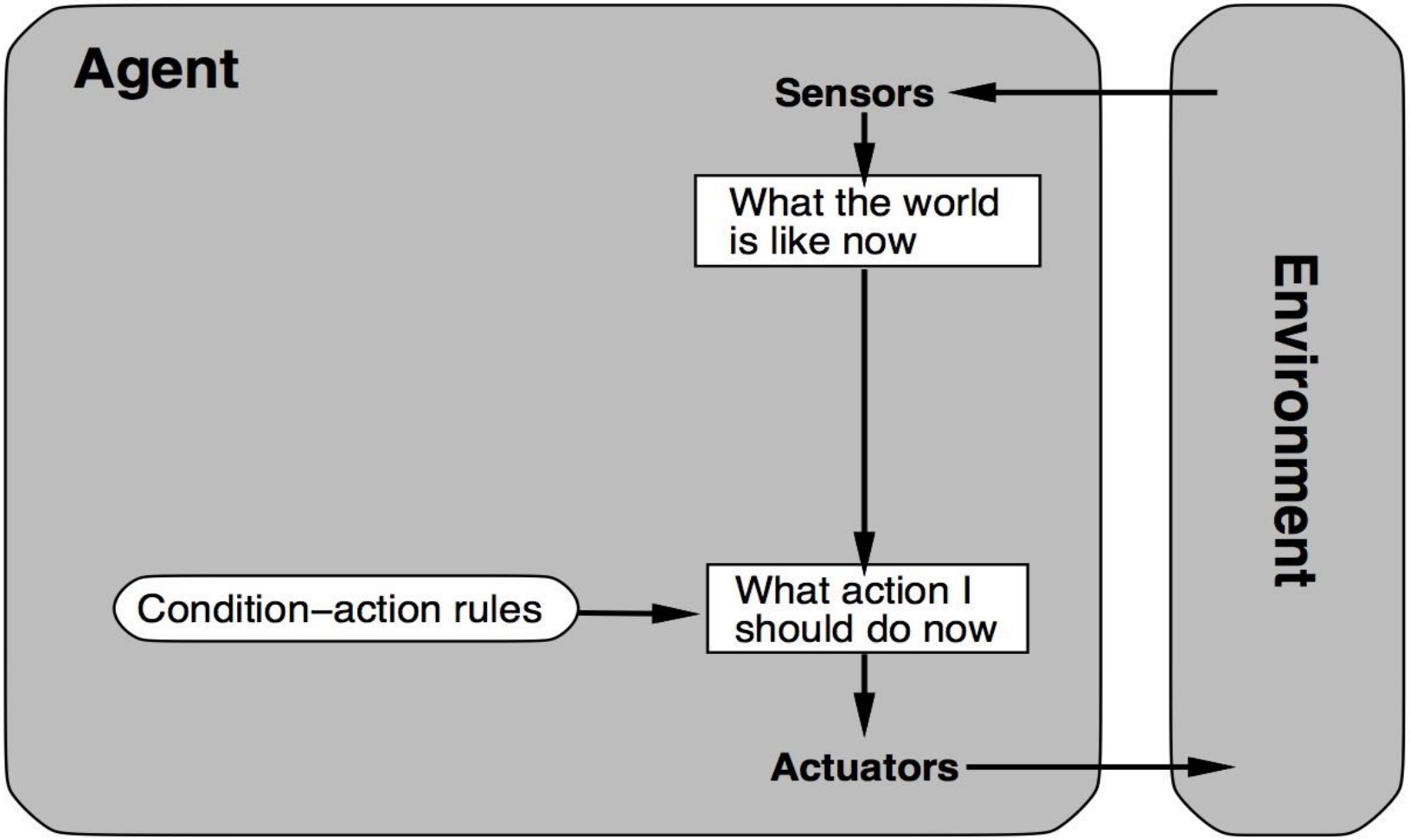
# Environment Examples

- Real World?
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  - Single-Agent vs. **Multi-Agent**

# Agent Types

- We will cover 4 basic types:
  - Reactive agents (reflex agents)
  - State-based agents
  - Goal-based agents
  - Utility-based agents
- They can all incorporate learning

# Reactive Agents

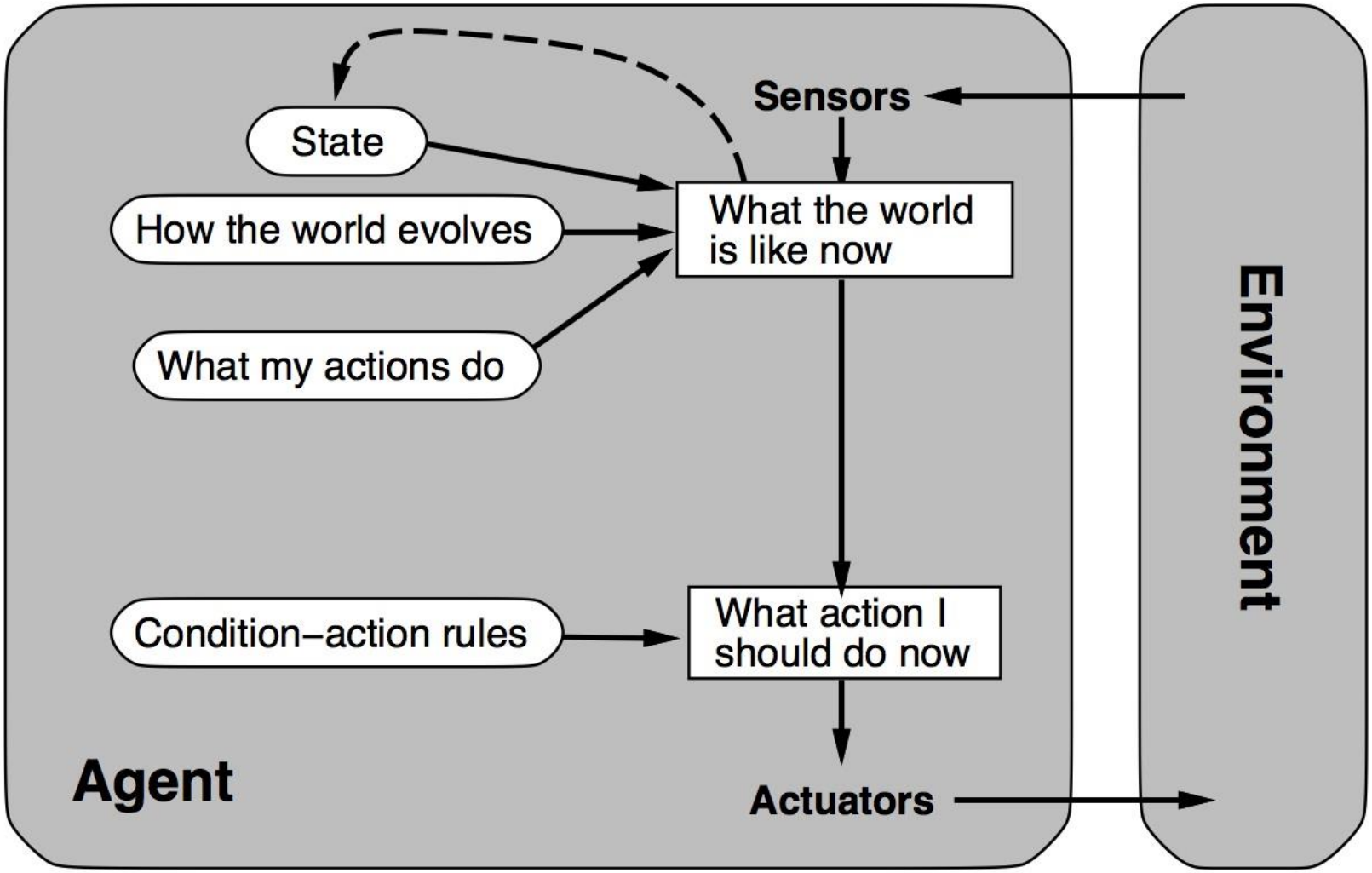


# Reactive Agent

- Finds a rule whose condition matches the current situation, as defined by percepts, and then does the action associated with the rule.
- Agent has no memory.

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

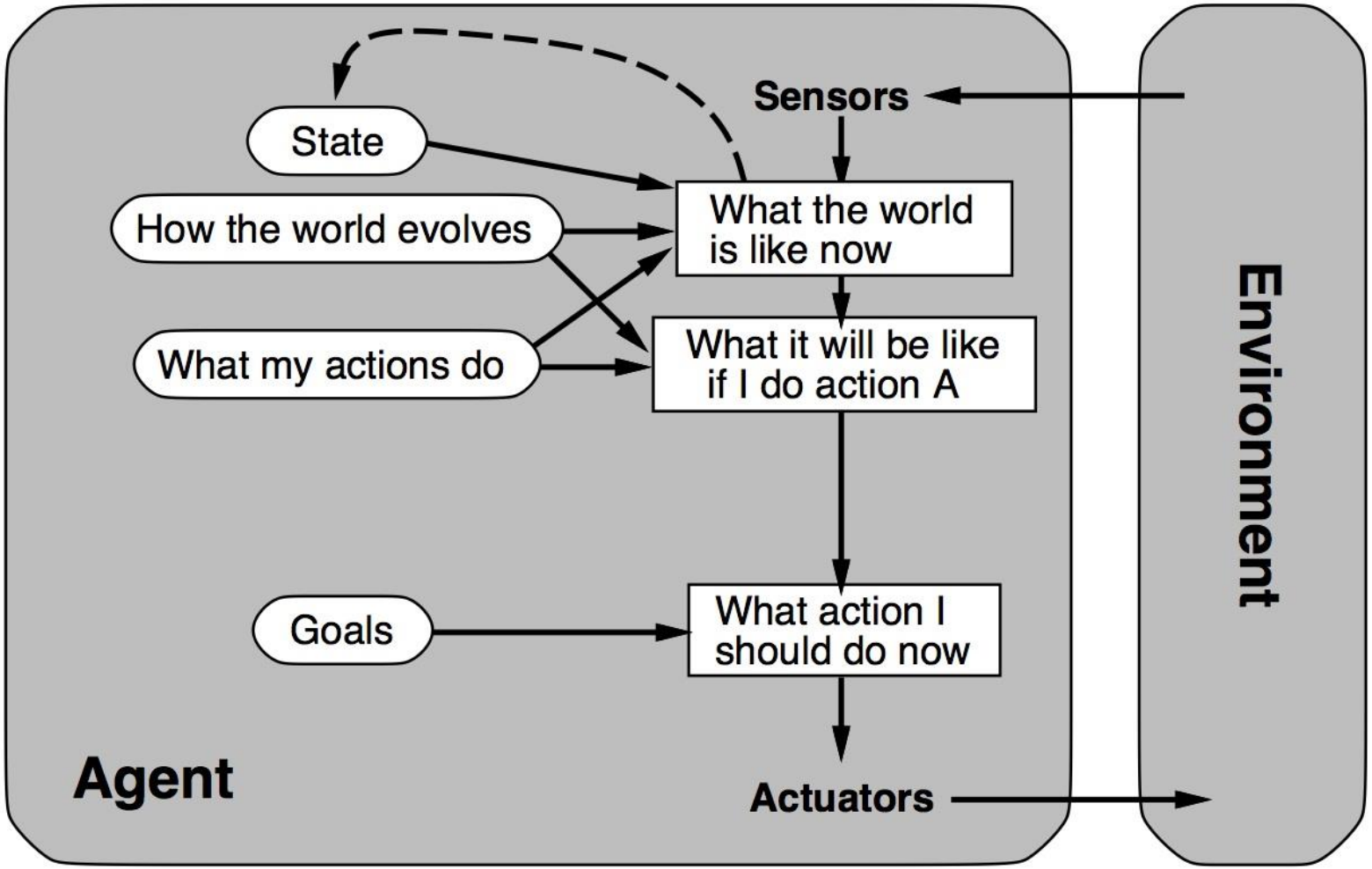
# State-based Agents



# State-based Agent

- Finds rule whose condition matches current situation, as defined by percepts and stored internal state, and does action associated with the rule.
- Internal state acts as a memory.

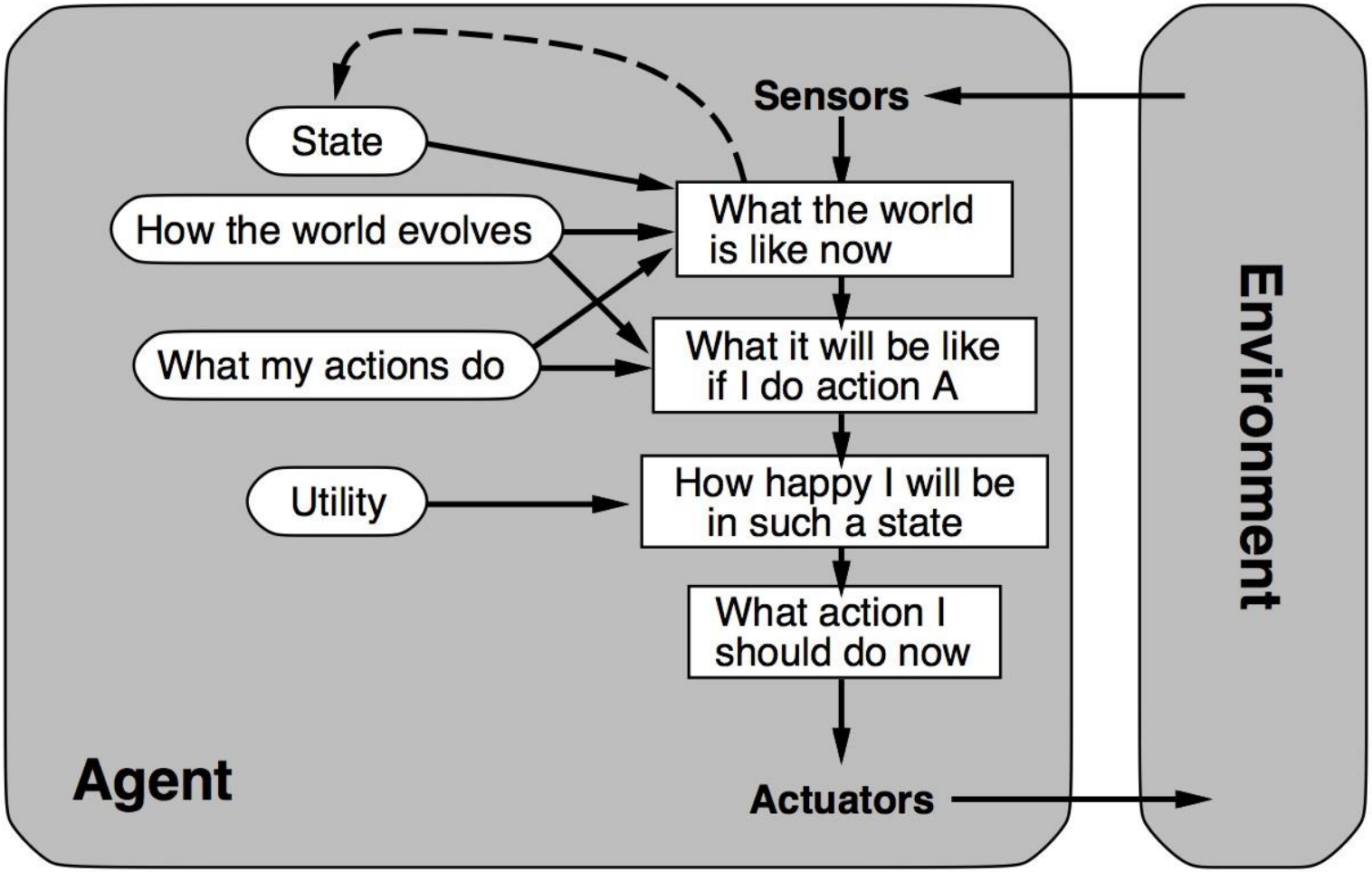
# Goal-based Agents



# Goal-based Agent

- Has explicit goal and when choosing an action, will select an action that achieves that goal.
- Requires some planning ("what will happen if I do action A?").

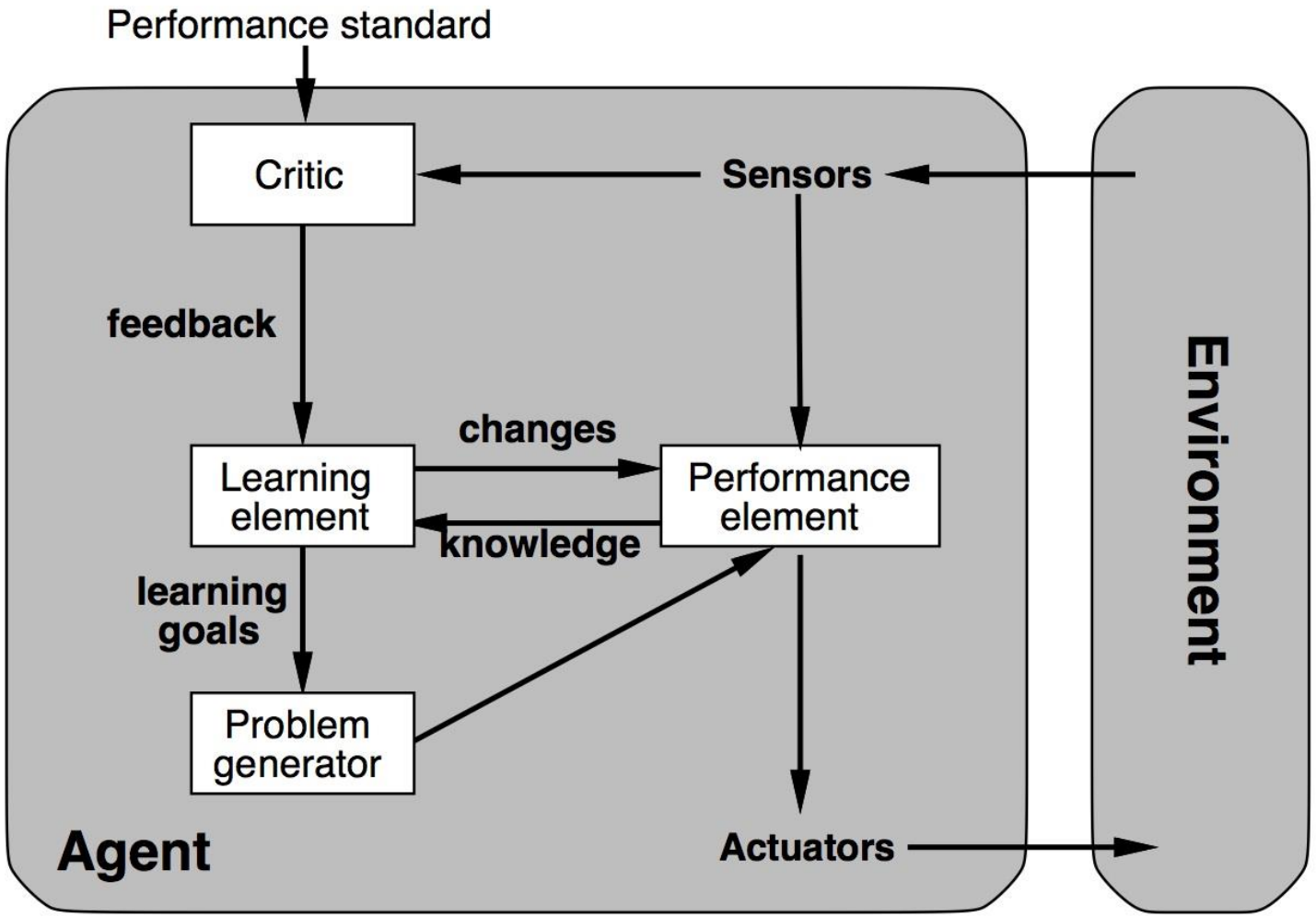
# Utility-based Agents



# Utility-based Agent

- Has utility function that maps a state onto real number that describes how well agent is performing.
- Not only achieves goals, but maximizes some utility measure.

# Learning Agents



# Summary

- Agents interact with environments through actuators and sensors
- The agent function maps percepts to actions
- A rational agent chooses the actions that maximize the performance measure
- PEAS describes components of rational agent design
- Environments can be categorized along several dimensions:
  - Observable? Deterministic? Sequential? Static? Discrete? Single-agent?
- Some basic agent architectures:
  - Reactive, State-based, Goal-based, Utility-based

# Looking Ahead…

- What we have covered so far…
  - What is AI?
  - What is a Rational Agent?
- In the next few lectures, we will cover…
  - How can rational agents solve problems?
  - How can rational agents represent knowledge?
  - How can rational agents learn?